



GE Transportation
a Wabtec company



Product Description

LXA and DP over Wi-Fi Mesh

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1. LOCOTROL® eXpanded Architecture

1.1 LOCOTROL® eXpanded Architecture (LXA) is Wabtec's next generation of Wabtec's LOCOTROL® platform. LXA may: a) enhance communications, and b) enhance connectivity for integration of certain key subsystems. LXA has the following capabilities.

1.2 Improved DP radio communications due to radio receive diversity

The standard LOCOTROL® DP system has two radios for DP communications. Hence, messages are received on only one radio at a time. The LXA DP system continuously listen on both radios and provides full radio diversity. Due to this feature, LXA reduces DP communication interruptions.

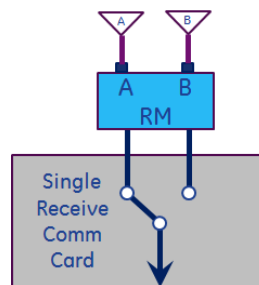


Figure 1- Standard DP radio communication – switch between radios

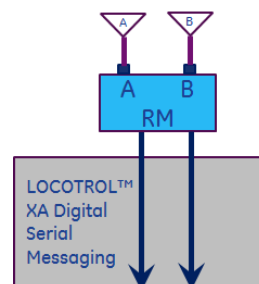


Figure 2- LOCOTROL® XA DP communication – full radio diversity

1.3 LXA Communication for the 272 Car Train

Although LXA provides a more robust comm solution capability, it alone will not enable the 272-car train to transit all of MRS' territory. LXA will need other enabling technologies like message repeaters, DP over IP communication technologies, and in some cases operational rules to effectively allow the 272-car train to transit through MRS' network.

1.4 Improved Hardware Platform

With the LXA platform, the DP and CCBII airbrake systems are more clearly partitioned. On the standard DP system, processor loading needs to be carefully managed, as the DP and CCBII software share the same processor. On the LXA system, the DP and CCBII systems are integrated (from a systems perspective) but physically reside separately on

their own circuit card with their own set of processor(s). Also, on the standard DP platform, on locos with CCBII, if a change is needed to the CCBII system software, then the DP software also has to be updated since the DP and CCBII system are tightly coupled. With LXA, the DP and CCBII systems are loosely coupled. If a change is needed to the CCBII system software, then the DP software does not have to be updated (an update will be required on a case by case basis based on the change being introduced).

In addition, the LXA platform contains a Dual processor with a main Application Processor and a Check Processor, while the standard DP system has a single processor. The Application Processor executes the DP application. The Check Processor can monitor the Application Processor's execution and takes action if it detects that the Application Processor is not responding appropriately.

Also, unlike standard DP, LXA has an ethernet port that facilitates easier connectivity to other systems on-board and offboard the locomotive.

1.5 Improved Logging

Both standard DP and LXA DP utilize "event driven" logging. The LXA DP system has increased "event driven" logging and has a larger storage capacity (5 MB) compared to standard DP (65 KB).

In addition, LXA DP logs all DP crew messages. The standard DP system is not able to record DP crew messages due to limited memory capacity and processor usage. LXA's DP crew message logging may provide valuable insights when troubleshooting incidents related to operator interactions with the DP system.

Alternatively, DP logs can be retrieved via the Ethernet port on the LXA system's processor module. Wabtec will provide the customers with the applicable information so that they can retrieve the log. Wabtec will work with our customers to determine a solution to remotely offboard the logs.

1.6 **LOCOTROL® DP Software Features**

The LOCOTROL® Kits includes the following software features to Wabtec's LOCOTROL® DP systems currently deployed on MRS's LOCOTROL® fielded fleets, as described in more detail below in this Section 2.5 (**Software Features**). All features below will be provided in a dual language mode, both English and Portuguese.

(a) Removal of Transition Penalties

Currently, while transitioning the train into and out of DP, a penalty is enforced. This feature eliminates that enforcement and instead requires a full Independent/Locomotive brake application during transition into and out of DP. No brake pipe pressure reduction is required.

To achieve Removal of Transition Penalties when entering DP requires a change to the DP system only. To achieve Removal of Transition Penalties when exiting DP requires a change to the DP system and air brake system. The scope of effort included in this proposal is for DP only changes. Customer must work with its air brake supplier (New York Air Brake) independently for

the air brake related change to fully Removal of Transition Penalty when exiting DP. If required, GET can be the technical/commercial prime and will subcontract with the air brake supplier. This effort would need to be quoted separately after GET gathers input from the air brake supplier.

(b) Automatic Remote Status Repeating and Hold off Communication Flash until One Retry

In an effort to conserve DP radio bandwidth, only DP command messages are currently repeated from the Lead to the tail-end Remote via the mid-train Remote (in a train with more than 1 Remote consist). In contrast, status messages from Remote consists are not repeated via the mid-train Remote consists to the Lead consist.

As train length and the distance between the Lead and tail-end Remotes increase, the ability for Remote status messages to reach the Lead is becoming more difficult, resulting in missed status messages and communication losses displayed on the operator screen.

In order to improve the communication performance, an updated communication scheme has been developed; enabling in-train repeating of DP status messages via the mid-train Remote DP consist. As part of this feature enhancement, the Comm. Loss indication is held off until one Retry has been initiated by the Lead locomotive.

(c) Incremental Linking/Unlinking

Throughout the course of a train trip, DP Remote locomotive consists may be required for a portion of the trip due to track grade and curvature. Currently, the link/unlink process must be completed to add/remove Remote consists during a train trip, which can take up to 30 minutes per occurrence.

The Incremental Link/Unlink feature provides the ability to incrementally link additional DP Remote consists to a linked DP train while avoiding the link/unlink process. Likewise, DP Remote consists can also be individually unlinked, thus avoiding the unlink/re-link process for the entire train.

(d) Enhanced Set-Out Mode

This enhancement provides the option to leave the Remote Brake Valve Cut-In while in SETOUT Mode. It allows the remote to maintain the pressure in the Brake Pipe avoiding an Emergency Application. All other SETOUT Mode Parameters remain the same.

(e) Enhanced Same/Opposite Selection and Display

With this feature the display will no longer defaults to the Same or Opposite Selection. Instead it forces the operator to choose the selection they prefer.

(f) Removal of Radio Fail Message during Linking

The automated Radio Link Test is performed during the DP Linking process. Currently, the test results are displayed to the operator. This feature will suppress the test results, so they are not shown on the screen but rather logged in the DP Event Log to be retrieved by maintenance personnel as needed.

(g) Poll Remotes for BP Test

During the Brake Pipe Test, the DP Lead will periodically poll the DP Remotes to verify they have seen flow. This allows for increased success for the brake pipe test in congested areas and areas where a single message may become corrupted.

(h) BP Test enhancement

During cold weather operations, with long lengths between locomotive consists, the BP Test can be challenging to pass. This BP Test Enhancement provides a much greater chance to pass the BP test while running longer trains in extremely cold temperatures.

(i) Lead/Remote Swap

Traversing through a y-switch or changing directions of a DP train due to a dead-end track requires the unlinking/relinking process, consuming time on the main line or yard. This DP Lead/Remote Reconfiguration feature has been developed to enable swapping of the Lead and Remote configuration in a DP train without requiring the Unlink/Re-link sequence. If MRS implements removal of transition penalties, this feature may only be useful if MRS intends to run multiple Remote locomotives. This feature is applicable to both the LOCOTROL® IPM fielded fleet and LOCOTROL® XA fielded fleet.

(j) CLID Recovery Enhancement

To help mitigate CLIDs resulting from “False” unexpected flow, primarily during train movement, Wabtec developed a software enhancement described below.

If CLID occurs under the following conditions, Remote will remain “armed” to automatically CUT-IN once radio communications is restored:

- ✓ Remoted was CUT-IN prior to CLID (i.e., CLID was due to detected Unexpected Flow)

- ✓ Remote Mode was in NORMAL mode prior to CLID.

If the following conditions are met following CLID, the Remote will automatically CUT-IN without requiring an Automatic Application/Release:

- ✓ Radio Communications is restored within 90 minutes (Extended Comm Loss)

- ✓ Lead commanding Automatic Brake RELEASE

✓ Operator commands Remote back to Normal within 10 minutes of being stopped.

✓ Operator is prompted including command to Normal mode may Cut-In Brake Valve.

If any of the above conditions are not met. The Remote will enforce normal CLID recovery interlocks prior to CUT-IN (may require Automatic Application/Release).

(k) DP over Wi-Fi Mesh

The LOCOTROL® XA DP system can send DP messages over the Rajant Wi-Fi Mesh IP network to supplement and increase overall DP radio communication beyond line-of-sight communications, supported by wayside Wi-Fi Mesh relay nodes. Throughout a DP trip, there may DP radio comm losses due to the topography of the territory. Depending on the territory, these comm losses can last several minutes and cause operational issues like a Comm Loss Idle Down (CLID) of the Remote or can cause the operator to panic and make a mistake like stopping the train or unlinking while in a comm loss.

Using this technology, the LOCOTROL® XA system establishes a communications link over the Wi-Fi Mesh IP network after the system is placed into DP Run Mode. Whilst LXA is communicating on DP radios, each LXA locomotives synchronizes to available wayside Wi-Fi Mesh nodes but backhauled together. The minimum RF performance and time to establish mesh communication is controlled by a LXA configurable mesh cost parameter. Once established, the LXA system will utilize the Wi-Fi mesh IP network for locomotive control to remotes and status reporting back to Lead. The “Comm Loss Indicator Light” will not be displayed during this time if there is a DP UHF Radio communication loss to make DP operations as seamless as possible. Once the Wi-Fi Mesh network is no longer viable, LXA will utilize the DP UHF Radio communications path for locomotive control until LXA recognizes it has good communication over the Wi-Fi Mesh network again. There will be diagnostics events within the system logs that will provide a complete picture of DP trips and the train’s interaction of switching into and out mesh IP nodes. This will allow MRS’ team to understand the overall health of onboard and offboard LXA DP over Wi-Fi Mesh functionality.

MRS currently runs through particularly arduous terrain with respect to their 272-car train makeup and loading. With this technology, MRS will have more robust Distributed Power communications, allowing them to ensure successful trips between their mine and port throughout the Steel Line. At this time, MRS will be deploying the Wi-Fi mesh network utilizing Rajant Hawk radios at 30dBm. These were chosen in line with Proof-of-Concept tests along sections of the Steel Line.